

Bounded Reduction: Removing the R -Dependence in Sharp Saint–Venant Stability (Route A, Plan 1, Building Block **BOUNDEDREDUCTION**)

Abstract

We upgrade the sharp Saint–Venant stability inequality from the bounded setting to the global setting:

$$E(\Omega) - E(B_1) \geq c_{\text{SV}}^{\text{glob}}(N) \mathcal{A}(\Omega)^2, \quad |\Omega| = |B_1|,$$

for every open set $\Omega \subset \mathbb{R}^N$ of finite measure. The input is the bounded-domain inequality $E(\Omega) - E(B_1) \geq c_{\text{SV}}(N, R) \mathcal{A}(\Omega)^2$ on $\{\Omega \subset B_R, |\Omega| = |B_1|\}$ produced by the sibling **SAINTVENANTASSEMBLY** block. The reduction is the constructive truncation argument of Brasco–De Philippis–Velichkov [1, Lemmas 5.1 and 5.3] (De Giorgi-type L^∞ iteration plus energy cut-off), together with the rearrangement-based suboptimal $\mathcal{A}^4$ bound of [1, Lemma 5.2]. No compactness, no sequential extraction, no contradiction step is introduced: every constant is an explicit composition of named dimensional quantities.

1 Notation and standing hypotheses

Throughout $N \geq 2$. All sets are open subsets of $\mathbb{R}^N$ of finite Lebesgue measure. We use the standard normalisations of the master document:

- $u_\Omega \in W_0^{1,2}(\Omega)$, $-\Delta u_\Omega = 1$, the torsion potential;
- $T(\Omega) = \int_\Omega u_\Omega \, dx$, the torsional rigidity;
- $E(\Omega) = -\frac{1}{2}T(\Omega) = \min_{v \in W_0^{1,2}(\Omega)} \left(\frac{1}{2} \int |\nabla v|^2 - \int v \right)$, the BDV Saint–Venant energy;
- $\omega_N := |B_1|$, $T_N := T(B_1) = \omega_N / (N(N+2))$;
- $\mathcal{A}(\Omega) := \inf_{x_0 \in \mathbb{R}^N} |\Omega \Delta (B_1 + x_0)| / \omega_N \in [0, 2)$ when $|\Omega| = \omega_N$, the Fraenkel asymmetry;
- the scale-invariant Saint–Venant deficit

$$D(\Omega) := E(\Omega) |\Omega|^{-(N+2)/N} - E(B_1) |B_1|^{-(N+2)/N}.$$

For $|\Omega| = \omega_N$ one has $D(\Omega) = \omega_N^{-(N+2)/N} (E(\Omega) - E(B_1)) \geq 0$.

The input from **SaintVenantAssembly**

Theorem 1.1 (Bounded sharp Saint–Venant; [3]). *For every $N \geq 2$ and $R \geq 2$ there exists $c_{\text{SV}}(N, R) > 0$ (explicit in upstream constants) such that*

$$E(\Omega) - E(B_1) \geq c_{\text{SV}}(N, R) \mathcal{A}(\Omega)^2 \tag{1}$$

for every open $\Omega \subset B_R$ with $|\Omega| = \omega_N$.

Two BDV black boxes

Lemma 1.2 ([1] Lemma 5.1: L^∞ tail estimate). *There is $C_7 = C_7(N) > 0$ such that for every open Ω with $|\Omega| = \omega_N$ and every $R \geq 1$,*

$$\|u_\Omega\|_{L^\infty(\Omega \setminus B_{R+1})} \leq C_7 |\Omega \setminus B_R|^{1/N}. \quad (2)$$

Reference. [1, Lemma 5.1], by De Giorgi iteration on the truncated test function $\varphi_k^2(u_\Omega - s_k)_+$. Constructive; no compactness. $\square$

Lemma 1.3 ([1] Lemma 5.2: suboptimal $\mathcal{A}^4$). *There is $C_8 = C_8(N) > 0$ such that every open Ω of finite measure satisfies*

$$D(\Omega) \geq \frac{1}{C_8} \mathcal{A}(\Omega)^4. \quad (3)$$

Reference. [1, Lemma 5.2]; equivalent (via $E = -1/(2\lambda_{2,1})$) to [2, Thm. 1] with $p = 2, q = 1$. Pure rearrangement plus Pólya–Szegő; no contradiction. $\square$

2 The BDV truncation lemma

The technical heart of the reduction is [1, Lemma 5.3]. It produces, out of a set Ω with small deficit, a *bounded* surrogate $\tilde{\Omega}$ with controlled diameter, controlled deficit, and controlled asymmetry loss.

Lemma 2.1 ([1] Lemma 5.3: bounded truncation). *There exist dimensional constants $C_9 = C_9(N) > 0$, $\delta = \delta(N) \in (0, 1]$ and $d = d(N) \geq 1$ such that for every open Ω with $|\Omega| = \omega_N$ and $D(\Omega) \leq \delta(N)$ there exists an open set $\tilde{\Omega}$ with the following properties:*

$$|\tilde{\Omega}| = \omega_N, \quad (4)$$

$$\text{diam}(\tilde{\Omega}) \leq d(N), \quad (5)$$

$$D(\tilde{\Omega}) \leq C_9 D(\Omega), \quad (6)$$

$$\mathcal{A}(\Omega) \leq \mathcal{A}(\tilde{\Omega}) + C_9 D(\Omega). \quad (7)$$

Reference. [1, Lemma 5.3]. The construction combines the L^∞ tail estimate (2) with the cut-off test function $\varphi_k u_\Omega$ in the Saint–Venant variational characterisation of $E(\Omega \cap B_{k+2})$, giving [1, (5.11)]

$$E(\Omega \cap B_{k+2}) \leq E(\Omega) + C b_k^{1+1/N}, \quad b_k := |\Omega \setminus B_k| / \omega_N.$$

Iterating with the Saint–Venant inequality on $\Omega \cap B_{k+2}$ produces an integer $K \leq K(N)$ with $b_{K+1} \leq 2\hat{C} D(\Omega)$. The set $\tilde{\Omega} := r^{-1}(\Omega \cap B_{K+3})$ with $r = (|\Omega \cap B_{K+3}| / \omega_N)^{1/N} \in [1 - \hat{C} D, 1]$ satisfies (4)–(7), the last estimate following from $|\Omega \Delta B_1(x_0)| \leq |(\Omega \cap B_{K+3}) \Delta \Omega| + r^N |\tilde{\Omega} \Delta B_1(x_0/r)| + \omega_N(1 - r^N)$. The construction uses only the cut-off identity, the Saint–Venant inequality on subsets, and the De Giorgi iteration of Lemma 1.2; no compactness, no minimising sequence, no contradiction step appears. $\square$

Remark 2.2 (Translation freedom). The diameter bound (5) is translation invariant. After translating, we may assume $\tilde{\Omega} \subset B_{d(N)}$. The asymmetry $\mathcal{A}$ and the deficit D are both translation invariant.

3 The far-deficit branch

Lemma 3.1 (Far-deficit quadratic bound). *For every open Ω with $|\Omega| = \omega_N$ and $D(\Omega) \geq \delta(N)/4$,*

$$E(\Omega) - E(B_1) \geq \omega_N^{(N+2)/N} \frac{\delta(N)}{16} \mathcal{A}(\Omega)^2. \quad (8)$$

Proof. $\mathcal{A}(\Omega) < 2$ gives $\mathcal{A}(\Omega)^2/4 < 1$, hence $D(\Omega) \geq \delta(N)/4 \geq (\delta(N)/16)\mathcal{A}(\Omega)^2$. Multiplying by $\omega_N^{(N+2)/N}$ and using $E - E(B_1) = \omega_N^{(N+2)/N} D$ gives (8). $\square$

4 The bounded-domain branch

Fix the dimensional radius

$$R_*(N) := \max\{d(N), 2\}, \quad (9)$$

so that $R_* \geq d(N)$ (Lemma 2.1 produces sets inside B_{R_*} after translation) and $R_* \geq 2$ (the hypothesis of Theorem 1.1).

Lemma 4.1 (Near-deficit transfer). *Set*

$$\tilde{c}_{\text{SV}}(N) := \omega_N^{-(N+2)/N} c_{\text{SV}}(N, R_*(N)) > 0. \quad (10)$$

For every open Ω with $|\Omega| = \omega_N$ and $D(\Omega) \leq \delta(N)$,

$$\mathcal{A}(\Omega)^2 \leq \frac{4C_9(N)}{\tilde{c}_{\text{SV}}(N)} D(\Omega) + 4C_9(N)^2 D(\Omega). \quad (11)$$

Proof. Apply Lemma 2.1 to obtain $\tilde{\Omega}$ satisfying (4)–(7). By Remark 2.2, assume $\tilde{\Omega} \subset B_{d(N)} \subset B_{R_*(N)}$. Apply Theorem 1.1 with $R = R_*(N)$ to $\tilde{\Omega}$:

$$E(\tilde{\Omega}) - E(B_1) \geq c_{\text{SV}}(N, R_*) \mathcal{A}(\tilde{\Omega})^2.$$

Dividing by $\omega_N^{(N+2)/N}$ and using $|\tilde{\Omega}| = \omega_N$,

$$D(\tilde{\Omega}) \geq \tilde{c}_{\text{SV}}(N) \mathcal{A}(\tilde{\Omega})^2. \quad (12)$$

Combining with (6), $\mathcal{A}(\tilde{\Omega})^2 \leq D(\tilde{\Omega})/\tilde{c}_{\text{SV}} \leq C_9 D(\Omega)/\tilde{c}_{\text{SV}}$, so $\mathcal{A}(\tilde{\Omega}) \leq (C_9 D(\Omega)/\tilde{c}_{\text{SV}})^{1/2}$. Plugging into (7) and squaring,

$$\begin{aligned} \mathcal{A}(\Omega)^2 &\leq (\mathcal{A}(\tilde{\Omega}) + C_9 D(\Omega))^2 \leq 2\mathcal{A}(\tilde{\Omega})^2 + 2C_9^2 D(\Omega)^2 \\ &\leq \frac{2C_9}{\tilde{c}_{\text{SV}}} D(\Omega) + 2C_9^2 D(\Omega)^2 \leq \frac{2C_9}{\tilde{c}_{\text{SV}}} D(\Omega) + 2C_9^2 \delta(N) D(\Omega), \end{aligned}$$

where we used $D(\Omega) \leq \delta(N) \leq 1$. Absorbing 2 into the constant 4 for convenience gives (11). $\square$

Corollary 4.2 (Near-deficit quadratic bound). *With*

$$c_{\text{near}}^{\text{glob}}(N) := \frac{\omega_N^{(N+2)/N}}{4C_9(N)(1/\tilde{c}_{\text{SV}}(N) + C_9(N))} > 0, \quad (13)$$

every open Ω with $|\Omega| = \omega_N$ and $D(\Omega) \leq \delta(N)$ satisfies

$$E(\Omega) - E(B_1) \geq c_{\text{near}}^{\text{glob}}(N) \mathcal{A}(\Omega)^2. \quad (14)$$

Proof. By (11), $D(\Omega) \geq \mathcal{A}(\Omega)^2/(4C_9/\tilde{c}_{\text{SV}} + 4C_9^2)$. Multiply by $\omega_N^{(N+2)/N}$. $\square$

5 The main gluing theorem

Theorem 5.1 (BOUNDEDREDUCTION: global sharp Saint–Venant stability). *With*

$$c_{\text{SV}}^{\text{glob}}(N) := \min\left\{c_{\text{near}}^{\text{glob}}(N), \omega_N^{(N+2)/N} \delta(N)/16\right\} > 0, \quad (15)$$

every open $\Omega \subset \mathbb{R}^N$ with $|\Omega| = \omega_N$ satisfies

$$E(\Omega) - E(B_1) \geq c_{\text{SV}}^{\text{glob}}(N) \mathcal{A}(\Omega)^2. \quad (16)$$

$c_{\text{SV}}^{\text{glob}}(N)$ depends only on the dimension N (the R -dependence inherited from SAINTVENANTASSEMBLY is internalised at $R = R_*(N)$).

Proof. *Case A* ($D(\Omega) \geq \delta(N)/4$): Lemma 3.1 gives $E - E(B_1) \geq \omega_N^{(N+2)/N} (\delta(N)/16) \mathcal{A}(\Omega)^2$.

Case B ($D(\Omega) \leq \delta(N)$, in particular when $D(\Omega) \leq \delta(N)/4$): Corollary 4.2 applies.

The two cases overlap on $\delta(N)/4 \leq D(\Omega) \leq \delta(N)$ and together cover $[0, +\infty)$. Either way (16) holds with $c_{\text{SV}}^{\text{glob}}(N)$ as in (15). $\square$

Remark 5.2 (Removal of the volume normalisation). (16) stated for $|\Omega| = \omega_N$ extends to arbitrary finite-measure Ω by scale invariance of D and $\mathcal{A}$: writing $\Omega_* := (\omega_N/|\Omega|)^{1/N} \Omega$ one has $|\Omega_*| = \omega_N$, $\mathcal{A}(\Omega_*) = \mathcal{A}(\Omega)$, $D(\Omega_*) = D(\Omega)$, and (16) for Ω_* reads $D(\Omega) \geq \omega_N^{-(N+2)/N} c_{\text{SV}}^{\text{glob}}(N) \mathcal{A}(\Omega)^2$. In particular the input to the Kohler–Jobin block holds without any boundedness assumption: at fixed volume $|\Omega| = |B^*| = \omega_N$, $T(B^*) - T(\Omega) = 2(E(\Omega) - E(B^*)) \geq 2c_{\text{SV}}^{\text{glob}}(N) \mathcal{A}(\Omega)^2$.

Remark 5.3 (Sharpness). The exponent 2 on $\mathcal{A}$ in (16) is sharp; it is saturated by smooth volume-preserving ellipsoidal perturbations of B_1 (see [1, §1.2]).

6 Interface with the chain

Upstream input. The block consumes the single number $c_{\text{SV}}(N, R_*(N))$ supplied by SAINTVENANTASSEMBLY at the dimensional radius $R_*(N) = \max\{d(N), 2\}$. Since $d(N)$ comes from Lemma 2.1 and is purely dimensional, so is $R_*(N)$.

Downstream output. The block emits $c_{\text{SV}}^{\text{glob}}(N)$ of (15). The Kohler–Jobin block takes $c_{\text{SV}}^{\text{glob}}(N)$ as input in place of $c_{\text{SV}}(N, R)$ and emits

$$c_{\text{FK}}(N) := \min\left\{\frac{4L_N c_{\text{SV}}^{\text{glob}}(N)}{(N+2)T_N}, \frac{L_N(2^{2/(N+2)}-1)}{4}\right\},$$

both arguments now functions of N alone. The final Faber–Krahn constant becomes $c_{\text{FK}} = c_{\text{FK}}(N)$ as desired.

7 Summary of constants

Symbol	Source	Meaning
ω_N	$ B_1 $	ball volume
T_N	$\omega_N/(N(N+2))$	torsion of B_1
$C_7(N)$	[1] L5.1	De Giorgi L^∞ tail constant
$C_8(N)$	[1] L5.2	suboptimal $\mathcal{A}^4$ constant
$C_9(N)$	[1] L5.3	truncation transfer constant
$\delta(N) \in (0, 1]$	[1] L5.3	truncation deficit threshold
$d(N) \geq 1$	[1] L5.3	truncation diameter bound
$c_{\text{SV}}(N, R)$	SVASSEMBLY	bounded SV constant
$R_*(N) := \max\{d(N), 2\}$	(9)	dimensional working radius
$\tilde{c}_{\text{SV}}(N) = \omega_N^{-(N+2)/N} c_{\text{SV}}(N, R_*)$	(10)	bounded SV in D -form
$c_{\text{near}}^{\text{glob}}(N)$	(13)	$\frac{\omega_N^{(N+2)/N}}{4C_9(1/\tilde{c}_{\text{SV}} + C_9)}$
$c_{\text{SV}}^{\text{glob}}(N)$	(15)	$\min\{c_{\text{near}}^{\text{glob}}, \omega_N^{(N+2)/N} \delta/16\}$

Where compactness might have appeared

- BDV Lemma 5.3 (truncation) is proved by *constructive* De Giorgi iteration plus a cut-off competitor: no minimising sequence, no Hausdorff limit, no contradiction. The integer $K \leq K(N)$ at which $b_K \leq 2\hat{C}D$ is reached by a geometric iteration [1, (5.12)–(5.14)], not by a compactness argument.
- BDV Lemma 5.1 (De Giorgi L^∞ tail) is constructive.
- BDV Lemma 5.2 ($\mathcal{A}^4$) is rearrangement-based.
- The gluing of the near and far branches is arithmetic.

References

- [1] L. Brasco, G. De Philippis, B. Velichkov, *Faber–Krahn inequalities in sharp quantitative form*, Duke Math. J. **164** (2015), 1777–1831 (arXiv:1306.0392); in particular Lemmas 5.1, 5.2, 5.3.
- [2] N. Fusco, F. Maggi, A. Pratelli, *Stability estimates for certain Faber–Krahn, isocapacitary and Cheeger inequalities*, Ann. Sc. Norm. Super. Pisa Cl. Sci. (5) **8** (2009), 51–71.
- [3] SAINTVENANTASSEMBLY, Plan 1 / Route A, companion file `SaintVenantAssembly.tex`.